

CHAPTER 5 · CONCEPT 3

Non-contact Forces

Forces that act from a distance, without touching — magnetic, electrostatic and gravitational.

Forces that act at a distance

Forces whose effect is felt **even when objects are not in contact** are **non-contact forces**. There are three to know.

Magnetic force

The force a magnet exerts on another magnet or a magnetic material is **magnetic force**. **Like poles repel** (N–N, S–S); **unlike poles attract** (N–S). It can act from a distance — so it is non-contact.

Electrostatic force

Rubbing certain materials builds up **static charges** on them — the object becomes **charged**. A charged object can attract light uncharged objects (paper bits) and other charges from a distance.

- There are two kinds of charge: **positive** and **negative**.
- **Like charges repel; unlike charges attract.**

Gravitational force (gravity)

The Earth pulls every object towards itself — the **gravitational force**, or **gravity**. It acts without contact, so it is non-contact, and it is **always attractive** (it only pulls, never pushes). That is why a thrown ball always falls back down.

Key contrast: magnetic and electrostatic forces can attract *or* repel; gravity is **always attractive**.

Where students lose marks

Trap 1 — Like repel, unlike attract. True for both magnetic poles and electric charges. Don't reverse it.

Trap 2 — Gravity is always attractive. Unlike magnetic/electrostatic forces, it never repels.

Trap 3 — All three act without contact. Magnetic, electrostatic and gravitational forces are non-contact forces.

Charging: both the rubbing and rubbed objects get charged, but with *opposite* kinds of charge.

Model answers — write them like this

Example A. Two balloons rubbed with wool move apart. Why?

Step 1: Both balloons get the same kind of static charge.

Step 2: Like charges repel.

∴ The similarly charged balloons repel and move apart.

Example B. Why does a thrown ball always come back down?

Step 1: The Earth exerts a gravitational pull on the ball.

Step 2: Gravity is always attractive, pulling the ball towards Earth.

∴ Gravity pulls it back to the ground.

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